Chathura Ratnayake

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Education

Zhejiang University — Zhejiang, China
Master of Engineering (Professional), Industrial Design Engineering — Dec 2026 (expected)

Tiangong University — Tianjin, China
Bachelor of Engineering, Chemical Engineering and Technology — Sep 2018 – Jun 2022
GPA: 3.41/4.0

Publications & Posters

Conference Proceedings (Peer-Reviewed)

1. **Ratnayake, A. R. M. C. D.** (2025). Designing In-Built Mini-Games for Chatbots: Opportunity to Improve User Engagement in Customer Service and Education. In R. Yamanishi, A. Formico, Y. Sugiura, & K. Matsumura (Eds.), *Entertainment Computing – ICEC 2025 IFIP TC 14 Workshops* (pp. 92–105). DOI ↗ — **Lecture Notes in Computer Science (LNCS)**, Springer

Journal Covers & Invited Editorials

1. **Ratnayake, A. R. M. C. D.** (2026). Editorial: Water Changes Lives! From Naïve Beginnings to Purposeful Pursuits. *ACS EST Water*, 6(2), 539–540. [Featured Cover Art Article]. DOI ↗ — **ACS EST Water**, American Chemical Society (ACS)

Conference Abstracts

1. Gunawardana, E. G. W., Gunathilake, D. R., **Ratnayake, C.**, Jeewarathne, K. R. A. D. U., Makehelwela, M., Weragoda, S. K., Gamage, C. D., & Weerasooriya, R. (2023, December). Importance of study on biofilm characterization within drinking water distribution networks. In *Book of Abstracts of the 14th International Conference on Sustainable Built Environment (ICSBE 2023)* (p. 184). ↗

Poster Presentations

1. **Ratnayake, C.**, Rodriguez, G., Madhumali, P.W., & Hossain, J.S. (2024, June). Nature Sip: Portable Pressure-Powered Filter for Rural and Remote Areas. *World Eco-Design Conference (WEDC)*, Vienna, Austria.

2. Jeewarathne, D., **Ratnayake, C.D.**, Mummullage, D., Godawattha, N., Feng, Y., & Mingxia, W. (2020). Oil & Water Clean-Comprehensive Liquid Zero Discharge Technology. *The 6th College Students Innovation and Entrepreneurship Competition*.

3. Nauchali, H.G.V., **Ratnayake, A.R.M.C.D.**, Jayasinghe, N.S., Makehelwala, M., & Weragoda, S.K. (2023). Comparative Study on the Influence of Mobile Phase pH on the Retention Time of MCPA, Captan, and Diuron in RP-HPLC Analysis. *Kandy Conference*, Kandy, Sri Lanka.

Research Experience

Ningbo Science and Technology Innovation Centre — Jan 2025 - Present, Ningbo - China

- Designed a portable, pressure-driven filtration system for developing regions; utilized Fusion 360 and 3D printing to prototype a low-cost solution expanding clean water accessibility. Presented as a poster in conference, Vienna, Austria.
- Prototyped an assistive fluid-delivery medical device using Arduino automation to

provide hands-free, user-friendly hydration for individuals with limited mobility.
- Developed human-machine interface design guidelines for educational systems, optimizing process workflows and system interaction efficiency.

Joint Research and Demonstration Centre for Water Technology Nov 2022 – Aug 2023, Kandy – Sri Lanka

- Assisted in screening *Escherichia coli* isolates and profiling antibiotic resistance genes across 14 drinking water treatment plant sources along the Mahaweli River.
- Evaluated the operational efficiency, water quality outcomes, and membrane fouling risks of reverse osmosis (RO) and nanofiltration (NF) plants desalinating groundwater in Sri Lanka's dry zone.
- Investigated the microbial diversity and persistence of antibiotic resistance genes (sul1) in drinking water distribution biofilms under tropical environmental conditions to assess public health risks in Sri Lanka.
- Assisted in evaluating nanofiltration membrane fouling mechanisms and cleaning protocols to enhance operational performance in rural water filtration plants.
- Fabricating low-cost durable graphite anodes for local adaptation of Electrodialysis.
- Operated HPLC and CHN instrumentation, developed new SOPs, and assisted with GC-MS and LC-MS analyses while performing routine wet lab sample preparation.

Mentoring, Teaching & Demonstration Experience

Graduate Student Mentor 2024, Zhejiang University

- Mentored incoming international graduate students to facilitate transition, adaptation, and university resource navigation.

Temporary Laboratory Demonstrator Jan 2023 - Aug 2023, Joint Research and Demonstration Centre for Water Technology,

- Conducted practical laboratory sessions on HPLC.
- Demonstrated wet lab techniques, and instrument calibration.
- Mentored students and evaluated laboratory coursework.

Professional Experience

Engineering Intern (Volunteer) Aug 2021 – Jan 2023, National Water Supply and Drainage Board

- Performed preventive maintenance, analytical instrument calibration, and troubleshooting for water purification systems.
- Developed technical drawings of system components using AutoCAD under the guidance of a Mechanical Engineer.

Awards

IDE Masters scholarship - UN Technology bank & Zhejiang University. Sep 2023

Tiangong University Merit Scholarship for International Talents Sep 2022

Skills

Laboratory Instrumentation: HPLC, LC-MS, CHNOS Analyzer, ICP-MS, GC-MS, portable field meters (pH/EC/TDS/DO), Chromeleon CDS, TraceFinder.

Programming, Modelling, Design & IoT: AutoCAD, Fusion 360, Adobe Suite, Python, R, MATLAB, Jupyter, GitHub, Arduino, microcontrollers, sensor integration.

Outreach, Leadership & Interests

- Maintained and troubleshot EDR, RO, and nanofiltration water purification systems, conducted village-level water quality testing, and trained local plant operators.
- Coordinated stakeholder workshops and field-based water sampling as an organizing committee member for an APN Nature-Based Solutions project. ↗
- Represented Sri Lanka as a **Junior National Baseball player** in international championships (**Chicago, IL, USA, 2011–2012**).
- Developed an open-source platform providing accessible website templates and visual communication tools specifically designed for graduate students, faculty, and research laboratories. ↗
- Drawing, Baseball, Chess, Creative Coding, Web and Graphic Design, Walking